UNIVERSITY OF WOLLONGONG IN DUBAI

CSIT302: Cybersecurity

Assessment 2: Report

Security Posture Evaluation Using the Cyber Kill Chain: LockBit Ransomware

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1. Team Members Contributions

Student Name	Student ID	Contribution (Summary)
Hriday Kumar	8890766	• Developed the technical threat scenario, including the organisational environment, attack path, and LockBit ransomware progression. • Mapped attacker activity to the Cyber Kill Chain and MITRE ATT&CK, using relevant technical indicators and evidence sources. • Assisted with proofreading, formatting, reference checking, and overall report consistency.
Sibi Parampil	8935166	• Contributed the proposed security controls and incident response content, focusing on prevention, detection, and response measures. • Helped align recommended controls with the identified attack stages and organisational weaknesses. • Assisted with proofreading, formatting, reference checking, and overall report consistency.
Muhammad Yahya	8258168	• Prepared the background and conceptual discussion, including key cybersecurity concepts relevant to the report topic. • Supported the explanation of the Cyber Kill Chain, security posture evaluation, and attacker movement within enterprise environments. • Assisted with proofreading, formatting, reference checking, and overall report consistency.
Mohamed Shafeen	8303988	• Evaluated the organisation's current security posture, controls, and gaps in relation to the LockBit attack path. • Developed the control-gap analysis and supporting summary table using technical indicators, logging evidence, and network/security weaknesses. • Assisted with proofreading, formatting, reference checking, and overall report consistency.
Abdirahman Mohamed	8690807	• Prepared the opening and closing parts of the report, including overall framing, context, and summary of findings. • Helped connect the report's technical analysis, key findings, and final conclusions into a coherent structure. • Assisted with proofreading, formatting, reference checking, and overall report consistency.

2. Introduction

Among the top challenges of cybersecurity, ransomware is the problem that organizations encounter. Ransomware attacks like LockBit have the potential to inflict significant financial losses and disruptions to the daily workflows because they attack employee computers, servers, and backup systems. The rationale behind the success of LockBit is that it relies on simple yet effective methods, such as spam emails, the theft of log-in credentials, and navigation inside the networks after gaining access. Once attackers have acquired the necessary control, they encrypt valuable systems and tend to steal valuable data to leverage organizations to pay a ransom.

The present report analyses a LockBit ransomware attack with the Cyber Kill Chain model. The Cyber Kill Chain breaks down an attack into various stages, starting with the first contact with a target and finishing with the eventual attack of systems and information. This is where it is convenient to refer to each of the stages separately to know how the attackers operate and where the existing security control mechanisms fail or are absent.

A case study is used to help in the analysis by taking a fictitious mid-sized company, ABC manufacturing Ltd. The company is a realistic architecture that consists of on-premise Active directory, remote access by VPNs, and restricted security monitoring. In the report, one can see how the lack of access control, endpoint protection, network segmentation, and logging could enable an attacker to migrate to another workstation that has been compromised to complete control of the network.

Lastly, the report pinpoints the main security gaps and proposes feasible steps to mitigate the risk and effects of ransomware.

3. Background and Key Concepts

3.1 Cyber Kill Chain Model

Cyber Kill Chain is a procedural roadmap invented by Lockheed Martin (2011) to present stages of the cyberattack process, starting with reconnaissance and ending with its ultimate effect. The Cyber Kill Chain is used to identify all areas where an intrusion should be detected or prevented by mapping the process that is stealthy to an attacker.

Lockheed Martin (2011) identifies seven phases that make up the model, which can be followed in ransomware attacks nowadays like Lockbit (Palo Alto Networks, 2024):

1. **Reconnaissance** - Attackers gather information about the target as much as they can get in order to discover vulnerabilities such as exposed services and portals to remote access.
2. **Weaponization** - Attackers create malicious code entries (or exploit kits), which are custom-made to address the vulnerability of the target.
3. **Delivery** - The ready-made ransomware elements are transmitted to the victim via phishing emails, system breach or even insider attacks.
4. **Exploit** - The attacker causes the execution by taking advantage of the vulnerabilities using the delivery package. The weak authentication and unpatched systems are often used by the attacker to execute a LockBit attack.

5. **Installation** - Ransomware programs are then installed on the host which has been compromised and is done to be persistent so that the attackers can have persistent access. This could be through the development of valid accounts or developing services.
6. **Command and Control (C2)** - This involves using backdoors that were made during the installation phase, it enables the attackers to establish communication with target system to organize encryption activity or rob data.
7. **Activities on Objectives** - At this point, files are mass encrypted and data is extracted by the attacker. This involves the theft of valuable information and planting of ransomware in Lockbit campaigns to get the highest extortion value.

3.2 Security Posture Evaluation

Security posture is the general quality, the maturity and efficiency of organizational cybersecurity defences in its technical, administrative and operational controls. It indicates the capacity of an organization to stop, identify, react and recuperate an occurrence of cyber assaults.

Security posture is not characterized by a number of controls, but rather a combination of a number of layers of protection. These consist of the perimeter protections (e.g. firewalls and network segmentation), endpoint security (e.g. antivirus and Endpoint Detection and Response (EDR) solutions), sound identity and access control (e.g. multi-factor authentication), centralized logging and monitoring, and inflexible vulnerability management procedures to see to it the vulnerable systems are patched as soon as possible.

Visibility and constant oversight are becoming more important in the modern security approach, especially in those environments where remote work, cloud use, and the Bring Your Own Device (BYOD) guidelines increase the conventional network perimeter. In accordance with the recommendations offered in NIST Cybersecurity Framework, the proper management of postures presupposes the corporations should recognize critical assets, secure them with the help of suitable protection, monitor suspicious sources, respond to any breaches, and provide some backup plans (National Institute of Standards and Technology, 2018).

3.3 Lateral Movement and Privilege Escalation

Lateral movement refers to the methods attackers use to move from one device to another after initial access. Rather than staying on a single compromised host, they pivot through the internal network to reach other systems, servers, and valuable assets, often using administrative protocols such as SMB (port 445), RDP (port 3389), and PowerShell remoting.

Privilege escalation is a process where attacker gains increment in his capacity of access in the environment he is attacking. This can be by either switching to a local administrator account or eventually switching to the Domain Administrator Platinum in Active Directory. It is often performed by attackers using credential dumping tools like Mimikatz, token impersonation or too permissive access control settings and permissions.

Lateral movement and privilege escalation are central to modern ransomware operations, not optional steps. Attackers usually delay ransomware deployment until they have mapped the network, identified critical systems, and obtained elevated privileges.

4. Technical Scenario and Threat Analysis

4.1 Organizational Network Scenario

To ground the threat analysis in a realistic context, this report models a fictional mid-size enterprise -ABC Manufacturing Ltd — with approximately 300 employees. The organization operates a standard on-premises Active Directory environment with the following infrastructure characteristics:

- Active Directory domain with a single Domain Controller (DC01)
- User LAN (192.168.10.0/24) and Server LAN (192.168.20.0/24) connected through a perimeter firewall
- Remote access via VPN (no Multi-Factor Authentication enforced)
- Shared file server and backup server hosted on the Server LAN
- Traditional signature-based antivirus on endpoints; no EDR solution deployed
- Windows Event Logging enabled locally; no centralized SIEM or log aggregation platform
- Outbound internet access broadly permitted, including unrestricted HTTPS

All SMB traffic (port 445) is permitted between the User LAN and the Server LAN, and administrative shares (C$, ADMIN$) are accessible from domain-joined workstations.

4.2 LockBit Attack Mapped to the Cyber Kill Chain

Table 4.2 – LockBit Kill Chain Mapping with Technical Indicators

Kill Chain Phase	LockBit Activity	Ports / Tools	Log Indicator	MITRE ATT&CK	Risk
Reconnaissance	Scans VPN portal & RDP endpoints, gathers employee emails via OSINT	TCP 443, 3389	Firewall deny logs, repeated connection attempts	T1595 - Active Scanning	High
Weaponization	Builds phishing email with malicious .docx or .exe payload	–	–	T1566.001 - Spearphishing Attachment	Medium
Delivery	Sends phishing email, exploits exposed RDP with stolen credentials	TCP 443, 3389	SMTP gateway logs, VPN auth logs	T1566 -Phishing, T1078 – Valid Accounts	High
Exploitation	User executes attachment; attacker runs PowerShell payload	TCP 445	Event ID 4624 (Logon), PowerShell Script Block logs	T1059.001 - PowerShell	High
Installation	Deploys persistence via scheduled task or registry run key	–	Event ID 4698 (Scheduled Task Created)	T1053.005 - Scheduled Task	High
C2	Beaconing to attacker server over HTTPS at regular intervals	TCP 443	DNS query logs, firewall outbound logs	T1071.001 -Web Protocols	High

Actions on Objectives	Encrypts files, deletes VSS backups, drops ransom note	–	Event ID 4663 (File access), VSS delete commands	T1486 - Data Encrypted for Impact	Critical

4.3 Realistic Attack Path and Asset Exposure

Table 4.3 – LockBit Attack Progression and Asset Exposure in ABC Manufacturing Ltd's Environment

Stage	Actor Action	Asset at Risk	Attack Path	Identity Impact
1. Initial Access	Phishing email / RDP credential abuse	Employee workstation (WS01)	Internet → Email Gateway → WS01 (User LAN 192.168.10.x)	Standard domain user account compromised
2. Foothold	PowerShell payload executes; scheduled task created for persistence	WS01 – User LAN	WS01 → Scheduled Task → Reverse Shell to C2 over TCP 443	User account executes attacker tooling in memory
3. Credential Dumping	Mimikatz dumps NTLM hashes directly from LSASS process	Domain Controller (DC01)	WS01 → LSASS process memory → NTLM hash captured	Domain Admin hash obtained (Pass-the-Hash attack ready)
4. Lateral Movement	Pass-the-Hash used to access SMB admin shares; pivot to servers	File Server (FS01) & Backup Server (BKUP01)	WS01 → TCP 445 → FS01 → BKUP01 (Server LAN 192.168.20.x)	Domain Admin account used to access all servers
5. Domain Compromise	Ransomware pushed via Group Policy Object to all domain machines	All endpoints, servers, DC01, BKUP01	DC01 → GPO → all domain-joined machines simultaneously	Full domain admin control; all assets and backups exposed

5. Security Posture, Controls, and Identified Gaps

5.1 Baseline Posture, Identity, and Endpoint/Server Gaps

ABC Manufacturing Ltd.'s baseline posture against the Heading 4 LockBit path includes a perimeter firewall, on-premises Active Directory (DC01), User LAN/Server LAN separation, VPN remote access, signature-based antivirus, local Windows event logging, and a backup server. These controls provide basic coverage but not layered detection or containment once a domain-joined workstation is compromised (The MITRE Corporation n.d.).

Key gaps in this baseline that directly support the Heading 4 attack path are:

- **VPN access that does not require MFA,** which would allow phished or stolen credentials to be reused to access the network, is in line with T1566/T1566.001 and T1078.
- **Only signature-based antivirus,** visibility of PowerShell abuse, dumping of credentials and scheduled-task persistence (T1059.001, T1003.001, T1053.005).
- **Simple privative-session separation provided by the scenario path** (credential theft on a user endpoint that ends up in domain-level compromise).
- **The situation does not have any evident incident escalation workflow,** which raises the likelihood that the suspicious VPN, host, or privilege activity is still present in subsequent stages of a kill-chain.

Together, these weaknesses allow LockBit to move from valid-account access to execution and persistence before effective containment occurs (The MITRE Corporation n.d.; CISA n.d.).

5.2 Network Architecture, Segmentation, and Exposure Gaps

ABC has logical segmentation (User LAN and Server LAN), but the effective access paths still support the lateral movement sequence in Heading 4.

- **The SMB (TCP/445) traffic is permitted between User LAN and Server LAN, administrative shares (C$, ADMIN$)** are accessible to domain-joined workstations and SMB/admin-share lateral movement and payload staging is immediately available (T1021.002).
- **Outbound HTTPS (TCP/443) is widely allowed** and C2 traffic is freely mixed with normal web traffic (T1071.001).
- **VPN/RDP exposure (443/3389)** enhances reconnaissance and targeting events, which is also in line with Active Scanning (T1595).

Once a user endpoint is compromised, the attacker can still reach higher-value systems with limited network resistance (The MITRE Corporation n.d.).

5.3 Logging, Monitoring, and Technical Evidence Gaps

Logging and monitoring are the largest visibility weakness because they affect detection across almost every phase after initial access. ABC is capable of producing useful evidence on the hosts of Windows, DC01, VPN/firewall and PowerShell, yet without centralized aggregation and correlation of events these events are in isolation and do not become usable incident timeline (Microsoft n.d.-a; Microsoft n.d.-b).

Even with the Heading 4, recognizable indicators such as 4624/4625, 4672, 4698 and 4663 would still be produced, and PowerShell logs would be emitted where enabled. Such artefacts are associated with credential abuse, persistence, privilege escalation, and ransomware execution, but are unlikely to be correlated over time to avert damage (Microsoft n.d.-a; Microsoft n.d.-b; Microsoft n.d.-c; Microsoft n.d.-d; The MITRE Corporation n.d.).

5.4 Backup, Recovery, and Vulnerability Handling Posture

ABC has a backup server, which is a useful baseline control. But in the Heading 4 direction the attacker gets to server infrastructure once they steal credential and move laterally using SMB which exposes backup systems once they have acquired elevated privileges. The phase of impact (encryption and shadow-copy deletion) are in line with T1486 and T1490, which means that it becomes less possible to recover and the risk of downtime is high should the backup infrastructure be accessible through the same path of compromise (The MITRE Corporation n.d.; CISA n.d.).

There is no formal vulnerability-management life-cycle (asset coverage, scan cadence, remediation tracking) of either of the two exposures (VPN/RDP) or high-risk internal routes (SMB/admin-share exposure) that the same path of intrusion would be viable over time.

5.5 Kill Chain Phase vs Existing Controls and Gaps

Table 5.5: Kill Chain Phase vs Existing Controls and Gaps

Kill Chain phase	Existing / baseline control in ABC Manufacturing	Specific gap	Resulting risk / how LockBit exploits it	Relevant ATT&CK technique IDs
Reconnaissance	Perimeter firewall; VPN/RDP exposed for business access	No monitoring of scanning/probing on VPN/RDP	443/3389 scanning goes undetected, enabling target profiling	T1595
Weaponization*	Basic email filtering (indirect control)	No advanced attachment inspection/sandboxing	Malicious lure/attachment reaches users	T1566.001
Delivery / Initial Access	Email gateway; VPN access; AD accounts	No MFA on VPN; weak phishing resistance	Stolen credentials enable foothold	T1566, T1566.001, T1078
Exploitation / Execution	Signature-based antivirus; local logs	No EDR / limited behavioural visibility	PowerShell execution proceeds with low detection	T1059.001
Installation / Persistence	Local Windows event logging	No central monitoring of persistence events	Scheduled-task persistence remains unnoticed	T1053.005
Command and Control (C2)	Outbound HTTPS allowed	Limited egress/DNS monitoring	HTTPS C2 blends with normal traffic	T1071.001
Lateral Movement / Privilege Escalation	AD domain; User/Server LAN split; SMB used operationally	SMB 445 + admin shares across segments; weak privilege separation	SMB movement to servers and DC paths	T1021.002, T1003.001, T1078
Actions on Objectives (Impact)	Backup server exists	Backups reachable via same admin path	Encryption + recovery disruption increase downtime	T1486, T1490

* Weaponization occurs on attacker infrastructure and is assessed here via ABC's pre-delivery controls.

6. Proposed Solutions and Incident Response

The analysis provided in Section 5 has identified three critical weaknesses that enable a LockBit attack: lack of MFA on VPN access, permissive SMB communication (TCP 445) that allows for lateral movement, and lack of centralized monitoring. The following controls target these gaps throughout the Cyber Kill Chain.

6.1 Preventive Controls

Phishing and Initial Access (T1566, T1078)
Implement good email filtering with attachment sandboxing and URL rewriting to decrease the arrival of malicious payload (T1566.001). Implement MFA on VPN and privileged accounts to avoid the reuse of credentials. Shut-off direct internet-facing RDP (TCP 3389) and set up account lockout policies to detect brute-force attempts (Event ID 4625).

Lateral Restriction of Movement (T1021.002)
Install VLAN based segmentation to separate the networks of User, Server and Backup. Block workstation-to-workstation SMB traffic and restrict administrative shares (C$, ADMIN$) to authorized administrators only. This restricts the pivoting between compromised endpoints to the Domain Controllers and file servers.

Endpoint Hardening and Credential Protection (T1059.001, T1003.001)
Deploy Endpoint Detection and Response (EDR) with behavioural monitoring. Enable LSASS protection (Credential Guard), PowerShell Script Block Logging, and enforce least-privilege with tiered administrative accounts. These controls reduce exploitation, credential dumping, and scheduled-task persistence (Event ID 4698).

Backup Resilience (T1486, T1490)
Implement immutable offline backups using a 3-2-1 strategy. Isolate backup systems from domain authentication and regularly test restoration procedures to ensure recovery from ransomware encryption.

6.2 Detective Controls

A centralized SIEM will then need to aggregate Windows Security Logs (4624, 4625, 4672, 4698, 4663), PowerShell logs, VPN logs, firewall and DNS records.

Example detection logic:

- Multiple failed VPN logins followed by successful login (credential abuse).
- Scheduled task creation coupled with encoded Power Shell execution.
- Abnormal SMB traffic from endpoints of users.
- Repetitive outbound beaconing of the unknown domains over HTTPS (T1071.001).

These measures enhance visibility during the Reconnaissance, Persistence and C2 phases.

6.3 Ransomware Incident Response Playbook

- Detection and Analysis: Look for signs like mass file changes (Event ID 4663), strange privilege assignments (4672), or new scheduled tasks (4698). Check the scope and the accounts that are affected.
- Containment: Right away, cut off infected systems, disable compromised accounts, and block malicious C2 traffic at the firewall.
- Eradication: Get rid of persistence mechanisms, reset privileged credentials, and reimage hosts that have been compromised.
- Recovery: Use verified offline backups to get systems back up and running and watch for reinfection before reconnecting to production networks.
- Lessons Learned: Update SIEM rules, make sure that segmentation is being enforced, and make phishing awareness training stronger.

6.4 Control-to-Kill-Chain Alignment

Proposed Control	**Kill Chain Phase Addressed**	**Risk Reduced**
Email sandboxing and Multi-Factor Authentication (MFA)	Weaponization, Initial Access	Prevents malicious attachments from executing and blocks reuse of stolen credentials
Network segmentation and SMB (TCP 445) restrictions	Lateral Movement	Limits pivoting between user endpoints, servers, and domain controllers
EDR deployment and credential protection (LSASS protection, least privilege)	Exploitation, Privilege Escalation	Detects PowerShell abuse and credential dumping attempts
Centralized SIEM with HTTPS egress monitoring	Command & Control, Persistence	Identifies beaconing behaviour and suspicious outbound traffic
Immutable and isolated backups	Actions on Objectives	Ensures recovery even if ransomware encrypts systems

7. Security Posture Measurement and Kill Chain Scorecard

Each kill chain phase is scored 1–5 based on how effectively controls prevent, detect, or mitigate the specific LockBit technique at that phase. 1 = Very Weak; 2 = Weak; 3 = Moderate; 4 = Strong; 5 = Very Strong.

Table 8 – Kill Chain Protection Scorecard: Before and After Proposed Controls

Kill Chain Phase	BEFORE			AFTER			What It Prevents
	Current Control	Score	Risk	Proposed Control	Score	Risk	Impact of Improvement
Reconnaissance	Perimeter firewall only; no scan detection	2	High	IDS (Suricata) at perimeter; scan alerting	3	Medium	Scanning now detected and alerted; limits attacker profiling
Weaponization	Basic email filter; no sandboxing	2	Medium	Email sandboxing; attachment detonation; URL rewriting	4	Low	Malicious payloads detonated before reaching user inbox
Delivery	No MFA on VPN; weak phishing protection	2	High	MFA on VPN; direct RDP disabled; account lockout policies	4	Low	Stolen credentials alone no longer grant network access
Exploitation	Signature AV only; no EDR	2	High	EDR with behavioural detection; PowerShell -Enc blocked; Credential Guard on LSASS	4	Low	LOLBin and fileless attacks caught; Mimikatz blocked from LSASS
Installation	Local Windows logs only; no SIEM alerting	2	High	SIEM with Event ID 4698 alerting; EDR task-creation monitoring	4	Low	Scheduled task persistence triggers immediate high-severity alert
Command & Control	Outbound HTTPS unrestricted; no DNS logs	1	High	DNS logging; forward proxy with SSL inspection; threat intel feed	3	Medium	Most C2 blocked; beaconing detected - residual risk: domain fronting
Privilege Escalation	No admin tiering; no privilege monitoring	2	Critical	Admin tiering (Tier 0/1/2); PAM solution; SIEM alert on Event ID 4672	4	Low	Pass-the-Hash path broken; Domain Admin creds not cached on workstations
Actions on Objectives	Backups domain-joined; VSS accessible	2	Critical	Immutable offline backups; BKUP01 off-domain; quarterly restore tests	5	Very Low	Recovery guaranteed regardless of encryption - ransom payment unnecessary

8. Conclusion and Reflection

This report demonstrated how a LockBit ransomware attack would impact a mid-sized organization by adhering to all the phases of the Cyber Kill Chain. Owing to the analysis, it was revealed that the vulnerabilities including the absence of multi-factor authentication in the entry of the VPN, unsecured SMB transmissions among other network divisions and a lack of monitoring make attackers prone to success. Once privileges have been raised via phishing or stolen credentials, attackers can proceed to move around the network elevating privileges and/or install ransomware with authorized tools such as Group Policy.

The review of the security posture demonstrated the need to go beyond minimum controls, including traditional antivirus and local event logs, to prevent contemporary ransomware. In the absence of EDR and centralized logs, the suspicious activity may be left undetected until the systems are encrypted. Despite the availability of the backup systems, the backup systems are under threat as well, when the attackers obtain access on the domain level, and make the recovery choices narrow.

Based on this analysis it is apparent that tighter, stricter, multi-layered security measures are needed. MFA, better network segmentation, EDR implementation, centralized monitoring, and segregated backups are some of the measures that can greatly mitigate the ransomware attacks and enhance readiness to respond to the incidents.

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10. Appendix

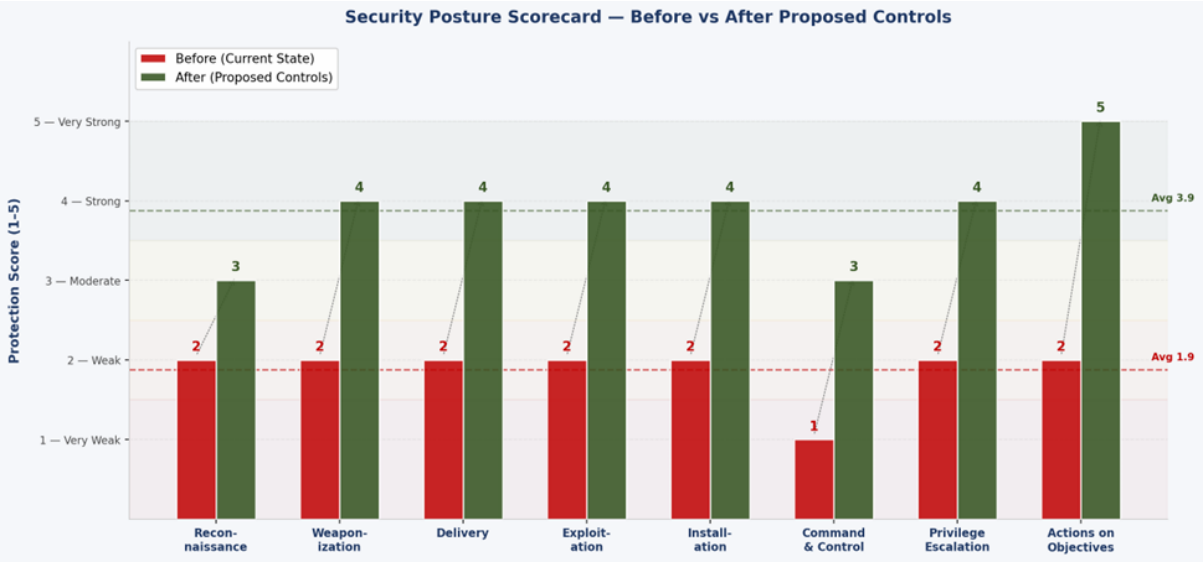


Figure 7: Security posture scorecard comparing ABC Manufacturing's control effectiveness before and after the proposed controls across each Cyber Kill Chain phase, scored on a 1–5 scale (1 = Very Weak, 5 = Very Strong)